Alfaisal Engineering Design Expo &
8th Boeing Engineering Student Competition

جامعة الفيصل
Alfaisal University

كلية الهندسة والحوسبة المتقدمة
College of Engineering and Advanced Computing

LLM-Augmented IoT Traffic Management

SE

SE-ALAWI-0042

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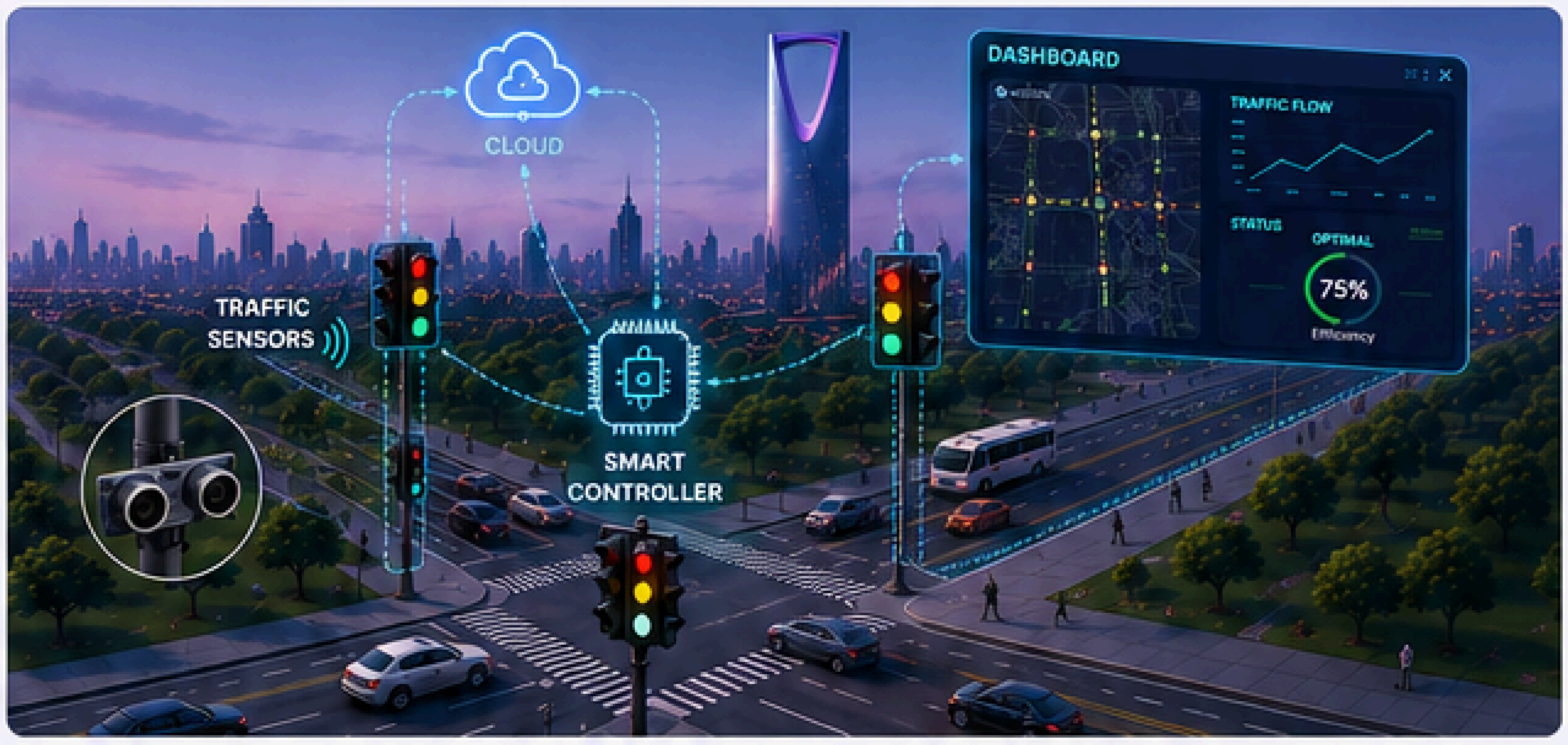
Urban traffic congestion costs the global economy hundreds of billions annually. Modern IoT-based adaptive controllers can respond to live vehicle queues, but they remain opaque to non-technical operators who must read raw telemetry streams and issue commands through proprietary interfaces.

We close this gap by adding a Large Language Model layer over a working ESP32/MQTT controller, enabling natural-language operation of an otherwise low-level system.



OBJECTIVES

- Build a working ESP32 adaptive traffic controller that reacts to live ultrasonic queue data.
- Add an LLM bridge that translates plain-language operator commands into structured MQTT control messages via tool-use.
- Deliver grounded conversational reporting on current and recent traffic state for operator situational awareness.

METHODOLOGY

Hardware
ESP32 + ultrasonic sensors + LED traffic lights

Communication
MQTT pub/sub over Wi-Fi

LLM Bridge
Python service with structured tool-use

Stack
Arduino · Python · HTML / JS

CAPABILITIES

- **Adaptive Timing** — green-light duration scales with the measured vehicle queue in each direction.
- **Natural-Language Control** — operators issue plain-English commands that the LLM converts into validated control messages.
- **Conversational Reporting** — operators can ask about current and recent traffic state and receive grounded summaries.

IMPACT

Improved Flow —
Up to 30% reduction in congestion

Operator Friendly —
Use natural language, not code

Scalable — Cloud-connected and city-ready

Smarter Cities — Data-driven, adaptive, sustainable

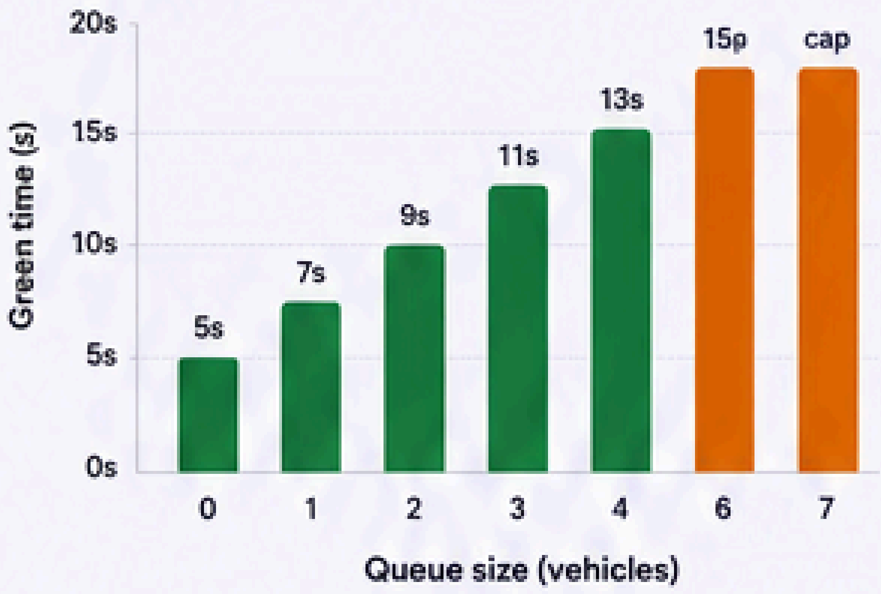
RESULTS

Fixed-Cycle vs Adaptive Timing

CONVENTIONAL TRAFFIC LIGHT
10s fixed
Same green time regardless of demand

OUR ADAPTIVE SYSTEM
5–15s queue-driven
Green time grows with detected vehicles

Green Time vs Queue Size



NATURAL-LANGUAGE CONTROL PIPELINE

1. OPERATOR SAYS
"give priority to west-east for 10 seconds"

→ LLM TOOL-USE

2. STRUCTURED COMMAND

```
{
  "direction": "WE",
  "action": "extend_green",
  "duration_ms": 10000
}
```

→ MQTT PUBLISH

3. LIGHTS RESPOND
WE green held for 10s, then resumes adaptive timing

CONCLUSION

The system pairs a working ESP32 traffic controller with an LLM bridge that turns plain-English operator intent into validated control messages and grounded conversational reports. Adaptive green time scales with measured queue depth and is bounded by a safe hard cap, while the language layer makes safety-relevant infrastructure usable without specialist training.

ESP32 / MQTT | LLM Tool-Use | Edge-Cloud Hybrid | Smart Cities | IoT

KEY TAKEAWAY

- Data-driven adaptive control reduces delays.
- Natural language makes systems accessible.
- Grounded reports improve operator awareness.
- Cloud connectivity enables smarter cities.

PROJECT REPOSITORY

Scan for GitHub Repository